Aritra Mandal

eraritramandal@gmail.com | Raipur, Chhattisgarh, 492004, IN | 8981308886
GitHub: https://github.com/NoxShadow17 | Portfolio: https://noxshadow17.github.io/Portfolio

PROFESSIONAL SUMMARY

Computer Science undergraduate (3rd year) specialized in engineering event-driven Generative AI systems, hybrid recommendation engines, and high-throughput RAG pipelines. Experienced in optimizing local vector databases, multi-agent orchestration, and deploying scalable full-stack applications with modern frameworks. Seeking work opportunities in applied AI and ML engineering.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, Java, C++, SQL
- **AI & Machine Learning:** scikit-learn, NumPy, Pandas, Local Embeddings (Xenova Transformers), RAG Pipelines, Context-Aware Prompt Engineering, Collaborative & Content Based Filtering
- **Web & Backend:** Next.js 14, Node.js, Express, FastAPI, Flask, REST APIs
- **Databases & Vector Search:** Supabase, PostgreSQL, MySQL
- **Tools & Developer Ecosystem:** Git, GitHub, Docker

PROJECTS

Advanced Movie Recommendation System

Github: https://github.com/NoxShadow17/Movie-Recommendation-system

Live: https://movie-recommendation-system-inky-alpha.vercel.app/

Python | FastAPI | React | SQLAlchemy | SQLite/PostgreSQL | Scikit-Learn | TMDB & YouTube APIs

- Engineered a hybrid recommendation engine combining user-based collaborative filtering (60% weight) via cosine similarity matrix scoring and content-based filtering (40% weight) utilizing Jaccard similarity across movie attributes.
- Optimized recommendation diversity by enforcing a 40% same-genre penalty cluster algorithm alongside personalized scoring boosts, caching results into a high-performance in memory service layer to avoid N+1 database queries.
- Built an interactive 13-page React frontend featuring real-time TMDB metadata streaming, synchronized watch parties, sentiment analysis on user reviews, and an embedded conversational AI assistant for interactive recommendation discovery.

NeuroAssist — AI-Driven Operating System for Neurodivergent Minds

Github: https://github.com/NoxShadow17/NeuroAssist-AI-Operating-System-for-Neurodivergent-Minds

Live: https://neuro-assist-ai-operating-system-fo.vercel.app/
Next.js | Node.js | Express | Supabase Auth & RLS | Groq SDK (Llama 3)

- Built a full-stack, distributed web application using a decoupled client-API architecture designed to reduce executive dysfunction and cognitive load through custom gamification systems and real-time XP ledgers.
- Developed a context-aware prompt engineering service using Groq SDK (Llama-3.3-70B and 3.1-8B) that dynamically rewrites system prompts on the fly based on individual user sensory profiles to deliver tailored micro-task breakdowns.
- Implemented strict tenant data isolation across the database schema using Supabase Row Level Security (RLS) policies (auth.uid() = user_id) combined with automated PL/pgSQL triggers for profile generation and daily streak management.

AVERT – Intelligent Cattle Collision Prevention System
Github: https://github.com/NoxShadow17/AVERT
Live: Hardware / Local Edge Deployment
Python | YOLOv8 | ONNX Runtime | WebSockets | C++ / ESP32 | OpenCV | Streamlit

- **Engineered a single-class object detection pipeline** by fine-tuning a YOLOv8 Small neural network on a cleansed, unified dataset optimized for harsh highway environments, achieving an 86.2% mAP@0.5, 91% peak recall, and a 0.82 Max F1-Score.
- **Optimized edge deployment and hardware communication** by exporting PyTorch weights into ONNX format to maximize inference FPS, alongside building an asynchronous, full-duplex WebSocket network using asyncio to bypass standard HTTP overhead.
- **Developed a decoupled AI-oT architecture** where the central Python server instantly pushes JSON telemetry to ESP32 microcontrollers for zero-latency LED roadside warnings, operating entirely independent of the real-time Streamlit surveillance dashboard to ensure life-safety triggers are unaffected by UI rendering lag.

Education:

- **Bhilai Institute of Technology (BIT), Durg, Chhattisgarh, India:** **2023-2027**
  - Bachelor of Technology in Computer Science & Engineering (3rd Year):
    - CGPA: 7.63/10
- **Delhi Public School (DPS), Durgapur, West Bengal, India:** **2022**
  - Higher Secondary Certificate(12th): 83.33%